

2026 ADA STANDARDS OF MEDICAL CARE IN DIABETES

Diagnostic Criteria, Glycemic Targets & Holistic Management

1. DIAGNOSTIC CRITERIA FOR TYPE 2 DIABETES

Diagnosis of diabetes requires confirmation by one of the following criteria (repeated on a second occasion):

- ? Fasting Plasma Glucose (FPG) ≥ 126 mg/dL (7.0 mmol/L) after an 8-hour fast.
- ? 2-Hour Plasma Glucose ≥ 200 mg/dL (11.1 mmol/L) during a 75-g oral glucose tolerance test (OGTT).
- ? Glycated Hemoglobin (HbA1c) $\geq 6.5\%$ (48 mmol/mol) using a certified NGSP assay.
- ? Random Plasma Glucose ≥ 200 mg/dL (11.1 mmol/L) in the presence of classic hyperglycemic symptoms (polyuria, polydipsia, unexplained weight loss).

2. PREDIABETES CATEGORIES

Impaired Fasting Glucose (FPG 100 to 125 mg/dL) or Impaired Glucose Tolerance (2-hr PG 140 to 199 mg/dL) or HbA1c 5.7% to 6.4% indicates high risk. Metformin therapy should be considered for prediabetes prevention.

3. INDIVIDUALIZED GLYCEMIC TARGETS

- ? Standard Target: HbA1c $< 7.0\%$ (53 mmol/mol) is recommended for most non-pregnant adults to reduce microvascular and macrovascular complications.
- ? Stringent Target ($< 6.5\%$): Appropriate for younger patients, short diabetes duration, and without significant cardiovascular disease, if achievable without clinically significant hypoglycemia.
- ? Relaxed Target ($< 8.0\%$): Appropriate for patients with severe hypoglycemia history, limited life expectancy, advanced microvascular or macrovascular complications, or extensive comorbid conditions.

4. LIFESTYLE INTERVENTIONS

Structured medical nutrition therapy (Mediterranean or plant-based diet) and ≥ 150 minutes per week of moderate-to-vigorous aerobic exercise combined with resistance training are foundational.

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First-Line Pharmacotherapy: Metformin Guidelines & Renal Precautions

5. FIRST-LINE ORAL AGENT: METFORMIN

Metformin (biguanide) remains the established foundational first-line agent for the management of type 2 diabetes in the absence of specific contraindications or end-organ complications.

MECHANISM OF ACTION:

Reduces hepatic gluconeogenesis, decreases intestinal absorption of glucose, and improves peripheral insulin sensitivity by activating AMP-activated protein kinase (AMPK).

DOSING & TITRATION PROTOCOL:

- ? Starting Dose: Metformin immediate-release 500 mg orally once or twice daily with morning and evening meals.
- ? Titration: Increase by 500 mg weekly to minimize gastrointestinal adverse effects (nausea, diarrhea, abdominal cramps).
- ? Maintenance Dose: 1000 mg orally twice daily (or Metformin Extended-Release 2000 mg once daily with the evening meal).
- ? Maximum Effective Dose: 2000 mg to 2550 mg per day.

6. RENAL FUNCTION MONITORING & METFORMIN SAFETY

Assess estimated Glomerular Filtration Rate (eGFR) prior to initiation and at least annually thereafter:

- ? eGFR ≥ 60 mL/min/1.73m²: Full dose safe (up to 2000-2550 mg/day). Monitor renal panel annually.
- ? eGFR 45 to 59 mL/min/1.73m²: Continued therapy permitted. Monitor renal function every 3 to 6 months.
- ? eGFR 30 to 44 mL/min/1.73m²: Dose reduction recommended. Maximum daily dose should not exceed 1000 mg daily. Do not initiate in treatment-naïve patients.
- ? eGFR < 30 mL/min/1.73m²: Strictly Contraindicated. Discontinue immediately due to the elevated risk of fatal Metformin-associated lactic acidosis (MALA).

VITAMIN B12 MONITORING:

Long-term Metformin use is associated with biochemical vitamin B12 deficiency. Periodic serum B12 testing is recommended every 2 to 3 years, particularly in patients presenting with peripheral neuropathy or anemia.

TABLE 1: METFORMIN eGFR DOSING PROTOCOL

eGFR ≥ 60 : Standard dose (up to 2000 mg daily)
eGFR 45 - 59: Maintain dose, monitor eGFR q6m
eGFR 30 - 44: Reduce dose (max 1000 mg daily)
eGFR < 30 : STRICTLY CONTRAINDICATED (Discontinue)

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Cardiorenal Protective Therapies: SGLT2 Inhibitors & GLP-1 Receptor Agonists

7. CARDIORENAL RISK-BASED SELECTION (INDEPENDENT OF HbA1c)

In patients with established Atherosclerotic Cardiovascular Disease (ASCVD), Heart Failure (HF), or Chronic Kidney Disease (CKD), SGLT2 inhibitors or GLP-1 receptor agonists with proven cardiovascular benefit are recommended as part of the glucose-lowering regimen, regardless of baseline HbA1c or Metformin use.

SODIUM-GLUCOSE COTRANSPORTER 2 (SGLT2) INHIBITORS:

Preferred in patients with Heart Failure (HFrEF or HFpEF) or CKD (eGFR 20 to 60 mL/min with UACR > 30 mg/g):

- ? Empagliflozin: 10 mg orally once daily; can titrate to 25 mg once daily. Proven reduction in cardiovascular death, HF hospitalizations, and CKD progression.
- ? Dapagliflozin: 10 mg orally once daily. Reduces HF hospitalizations and slows renal function decline.
- ? Adverse Effects: Mycotic genital infections, euglycemic diabetic ketoacidosis (euDKA), volume depletion.

GLUCAGON-LIKE PEPTIDE-1 (GLP-1) RECEPTOR AGONISTS:

Preferred in patients with established ASCVD (prior MI, stroke, revascularization) or high cardiovascular risk, and where weight reduction is a primary therapeutic goal:

- ? Semaglutide: Subcutaneous 0.25 mg weekly for 4 weeks (initiation dose, not effective for glycemic control), then increase to 0.5 mg weekly. If needed after 4 weeks, titrate to 1.0 mg weekly, up to maximum 2.0 mg weekly.
- ? Dulaglutide: 0.75 mg subcutaneously once weekly; titrate to 1.5 mg up to 4.5 mg weekly.
- ? Dual GIP/GLP-1 Agonist (Tirzepatide): 2.5 mg weekly for 4 weeks, titrate by 2.5 mg increments every 4 weeks to target dose of 5 mg, 10 mg, or maximum 15 mg weekly.
- ? Adverse Effects: Nausea, vomiting, diarrhea, pancreatitis risk. Contraindicated in personal or family history of medullary thyroid carcinoma (MTC) or MEN2 syndrome.

TABLE 2: CARDIORENAL AGENT PREFERENCES

Heart Failure (HFrEF/HFpEF): SGLT2i preferred
(Empagliflozin 10mg / Dapagliflozin 10mg)
Chronic Kidney Disease: SGLT2i (eGFR≥20) or GLP-1 RA if
SGLT2i contraindicated
Established ASCVD: GLP-1 RA or SGLT2i with proven benefit
Weight Loss Priority: Tirzepatide or Semaglutide

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Basal Insulin Initiation, Titration & Hypoglycemia Management

8. INSULIN THERAPY IN TYPE 2 DIABETES

When oral agents and GLP-1 RAs fail to achieve glycemic targets, or in symptomatic severe hyperglycemia (HbA1c > 10% or blood glucose $\geq$ 300 mg/dL with catabolic features), basal insulin should be initiated.

BASAL INSULIN PROTOCOL:

? Agents: Insulin Glargine U-100/U-300, Insulin Degludec U-100/U-200, or Insulin Detemir.

? Starting Dose: 10 units once daily at the same time each day, or weight-based 0.1 to 0.2 units/kg/day.

? Titration Algorithm: Self-titrate by increasing dose by 2 units every 3 days until fasting plasma glucose reaches target (80 to 130 mg/dL) without hypoglycemia.

? If fasting glucose < 70 mg/dL: Decrease basal dose by 10% to 20% immediately.

9. HYPOGLYCEMIA DEFINITIONS & MANAGEMENT

? Level 1: Glucose 54 to 69 mg/dL (Alert value).

? Level 2: Glucose < 54 mg/dL (Clinically significant).

? Level 3: Severe hypoglycemia accompanied by cognitive impairment requiring external assistance.

THE 'RULE OF 15' PROTOCOL:

1. Ingest 15 to 20 grams of fast-acting glucose (e.g., 4 glucose tablets, 1/2 cup fruit juice, or 1 tube glucose gel).

2. Recheck blood glucose level in exactly 15 minutes.

3. If blood glucose remains < 70 mg/dL, repeat 15 grams of glucose.

4. Once glucose normalizes, consume a meal or snack containing complex carbohydrates and protein to prevent recurrence.